

Tracing Software Defined Network Packets with Wireshark

This article describes the use of Wireshark for tracing software defined network (SDN) packets. Wireshark is a network packet analyser that enables the dissection of captured packet data for detailed information. This tool is widely used by network administrators to troubleshoot various issues.



Traditional methods of implementing communication technologies are usually based on hardware, which limits the scalability of the network. In spite of their extensive adoption, traditional networks are complex by nature and are difficult to manage. A traditional networking device packages the control plane and the data plane, which reduces the flexibility of the networking infrastructure. Software defined networking (SDN) is an emerging networking standard that aims to address the shortcomings of traditional networks.

SDN and its advantages over traditional networks

SDN is proposed as a standard to solve the issues in traditional networks, and provides flexibility, scalability, low cost and low power consumption.

Due to the increasing demand for a large number of devices in the IoT environment, a programmable model like SDN is required to maintain quality of service (QoS). The main purpose of SDN is to separate the control and forwarding functions of the network, which are the control plane and the data plane, respectively. Figure 1 shows the basic SDN architecture with three planes. The data plane can consist of multiple IoT or networking devices that generate the data. The control plane acts as a brain for the core network due to its centralised view. The SDN controller and the network devices can communicate with each other through the southbound API. An application plan can consist of various business applications.

A software defined network offers centralised controller and network management. One of its biggest advantages is that it permits the design of a framework that supports demanding applications like Big Data and virtualisation. Some of the benefits of software defined

- Centralised network management
- Greater scalability
- Lower operational costs
- Security
- Reduced hardware dependability

Why Wireshark is needed

Wireshark is a network packet analyser that allows dissecting the captured packet data for detailed information. It is widely used by network administrators to monitor how the network is working. Some of the uses of this tool are listed below.

- Network administrators use it to troubleshoot network issues.
 - Programmers use it to debug protocol implementations.
 - Network security administrators use it to inspect various security issues.
 - It is used to learn about the internal working of network protocols.
 - Quality assessment engineers use it to verify network applications.
- The various features of Wireshark are:
- Captures the live packets data from a network interface.
 - Can import packets from text files with hex dumps of packet data.
 - Creates various statistics on the packets captured.
 - Saves the packet data captured from a network interface.
 - Exports the packets in a number of capture file formats.
 - Searches and filters packets on various criteria.
 - Displays packets with different colours based on filters.

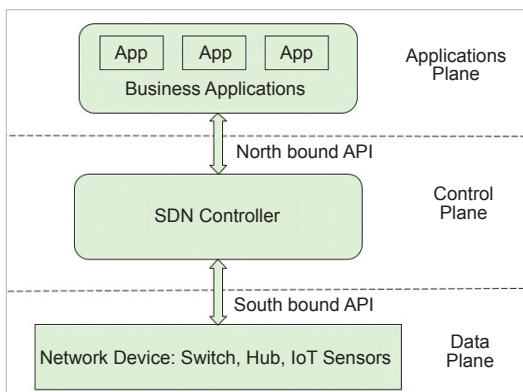


Figure 1: SDN architecture

networking are: